

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457717.75178 +/- 0.00033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.015
Transit depth (Rp/Rs)^2	1.88 +/- 0.41 [%]
Orbital Inclination (inc)	86.79 +/- 1.09
Airmass coefficient 1 (a1)	1.142 +/- 0.011
Airmass coefficient 2 (a2)	-0.1298 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	None
Optimal Aperture	4.56
Optimal Annulus	11.96
Transit Duration (day)	0.095 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.015 vs 0.1178 (deviation 1.22σ)
Duration fit vs published	0.095 d vs 0.125 d (deviation 25.9σ)
Mid-time O–C	-1571.0 min
Depth SNR	8.7
Residual scatter	0.96 %

Light curve

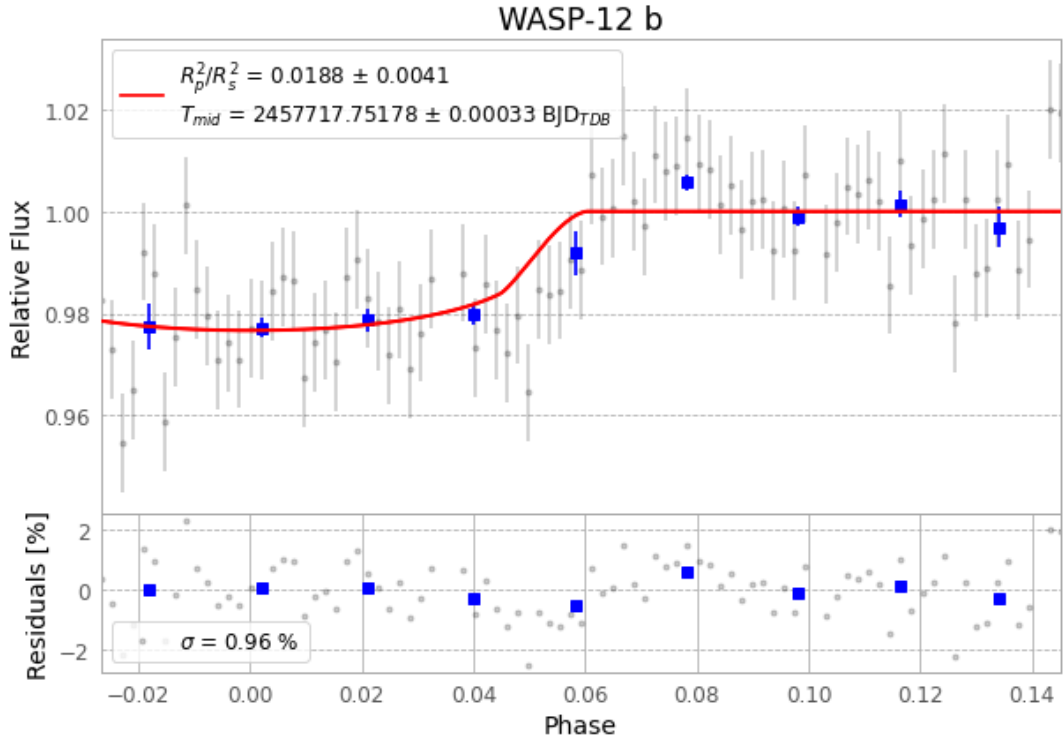


Figure 1 — FinalLightCurve_WASP-12 b_2016-11-24.png (SHA-256 8a0aac03c6ab52c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50d5e90fe0f085e5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), WASP-12161125051212.FITS → WASP-12161125094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161126.log (b0298ed56726...), FinalLightCurve_WASP-12 b_2016-11-24.png (8a0aac03c6ab...), FinalLightCurve_WASP-12 b_2016-11-24.csv (b69a1e70424d...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:42Z.